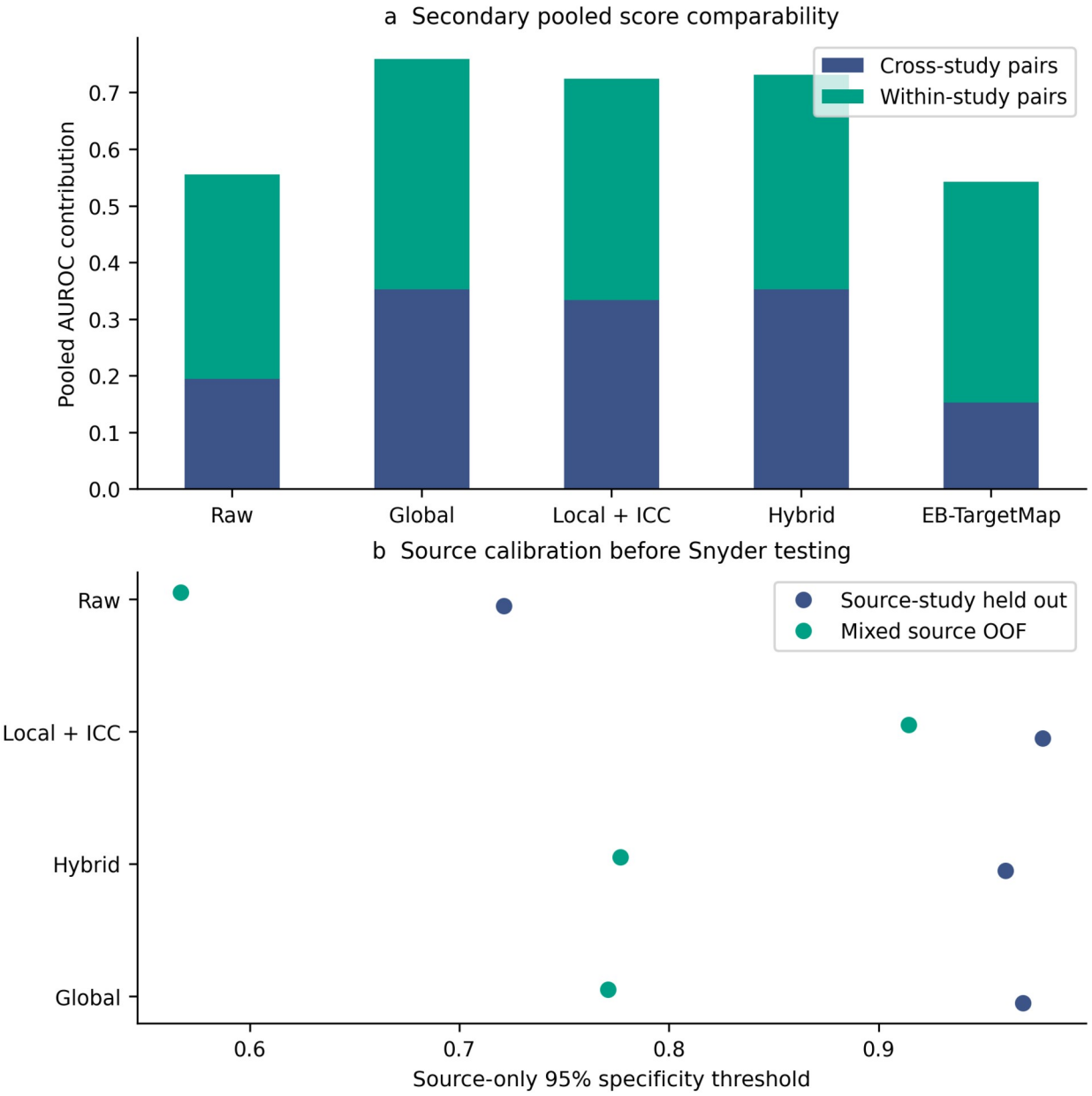
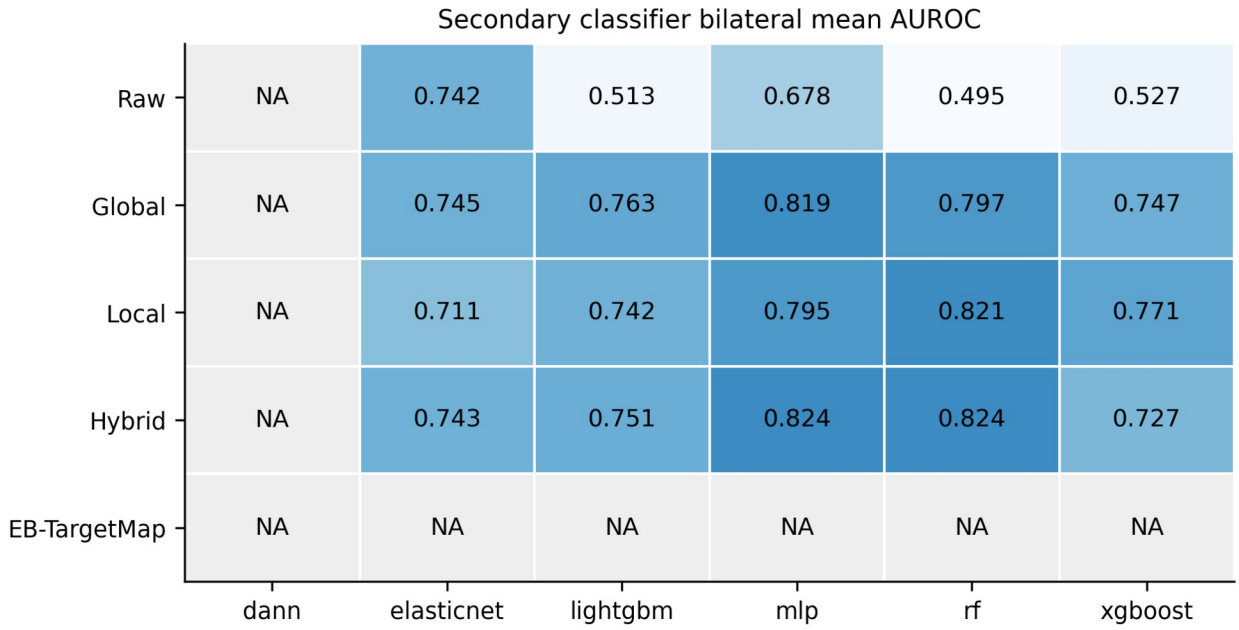
Supplementary material

Scale Trade-offs in Within-Sample Self-Normalization for Cross-Study Generalization in Public cfDNA Cohorts



Supplementary Figure S5. Secondary score comparability and source calibration. (a) Within-study and cross-study pair contributions to pooled Combined LOCO AUROC. (b) Mixed-source OOF versus source-study-held-out thresholds fitted before testing Snyder. Pooled discrimination is a cross-model score diagnostic; the primary performance evidence is direction-specific.



Supplementary Figure S6. Secondary classifier comparisons. Bilateral mean target AUROC by representation and fixed classifier. Gray cells denote unidentifiable EB-TargetMap/DANN configurations. Local uses its Local_LSN branch in the primary one-source design. Full random and study-held-out metrics are in Supplementary Table S6.

Statistical table guide

Supplementary Table S1: Primary_BH_Hypothesis_Family.csv. All 38 primary hypotheses, models, target studies, DeLong p and BH q values. The complete family remains the historical prespecified family.

Supplementary Table S9: Threshold_Calibration_Comparison.csv. Mixed-source OOF and source-study-held-out calibration, source threshold intervals and propagated operating points. Missing one-source EB-TargetMap cells are explicit.

Supplementary Table S3: Primary_Hybrid_Global_Delta.csv. Paired Hybrid−Global ΔAUROC and 1000 diagnosis-stratified patient-bootstrap intervals.

Supplementary Table S4: Depth_CancerType_Sensitivity.csv. Post-hoc exploratory sensitivity analyses of the shared-depth interval and GI-versus-HCC restriction, using the same public cohorts with altered target size/composition; not independent external validation or confirmatory evidence.

Supplementary Table S5: Chromosome_Block_Associations.csv. Nine genomic block tests with full block bootstrap intervals and block-wild BH correction. PERMDISP_Dispersion.csv gives the four separate dispersion tests.

Supplementary Table S6: Table3_Benchmark_Metrics_Full_Revision.csv. Complete fixed-classifier random and study-held-out metrics, with current method names and explicit unidentifiable cells.

Supplementary Table S7: Anchor_Distribution_Summary.csv. Median, Q1, Q3, IQR, full range and threshold-crossing rates. Anchor_All_Predictions.csv gives all repeated observations and Anchor_Real_Batch_Members.csv supplies patient identities for reproducibility.

Supplementary Table S8: Table2_Study_Depth_Fragments_Composition.csv. Exact depth/fragment ranges and cancer/stage categories for all sources. Source stage codes 0 and X remain as supplied.

Supplementary Table S2. Software versions.

package	version
numpy	2.2.6
pandas	2.3.3
pyarrow	24.0.0
scipy	1.15.3
scikit-learn	1.7.2
statsmodels	0.15.0
torch	2.5.1+cu121
xgboost	2.1.4
lightgbm	4.6.0
umap-learn	0.5.12
matplotlib	3.10.9
seaborn	0.13.2
requests	2.34.2
Python	3.10.10

Supplementary Table S9. Source threshold calibration (primary).

Target	Method	Source threshold [95% CI]	Target specificity	Propagated specificity [95% CI]
Cristiano	Global	0.629 [0.407, 0.744]	0.660	[0.512, 0.737]
Cristiano	Hybrid	0.559 [0.334, 0.617]	0.632	[0.517, 0.675]
Cristiano	Local LSN	0.671 [0.517, 0.805]	0.512	[0.445, 0.632]
Cristiano	Raw	0.695 [0.679, 0.709]	0.450	[0.105, 0.770]
Jiang	Global	0.729 [0.576, 0.881]	0.406	[0.281, 0.594]
Jiang	Hybrid	0.764 [0.624, 0.883]	0.531	[0.406, 0.719]
Jiang	Local LSN	0.866 [0.820, 0.915]	0.812	[0.750, 0.812]
Jiang	Raw	0.537 [0.506, 0.570]	1.000	[1.000, 1.000]